

## A2 Notes P-6 Binomial Probability

A set of  $n$  trials is called a **binomial experiment** if the following are true:

1. A **fixed number** of trials,  $n$ .
2. The **trials**(attempts) are **independent**
3. Each trial has **only 2 outcomes**; **success** ( $p$ ) and **failure** ( $q$ )

Note!  $p + q = 1$  (all probabilities of a situation must total to 1 (think 100%))  
So,  $q = 1 - p$

4. The **probability of successful outcomes,  $p$** , is the same number for each trial.

If  $n$  number of independent repeated trials, each with a probability of success  $p$  and a probability of failure  $q$  then the binomial probability of  $r$  number of successes in the  $n$  number of trials is found by:

$$P(r) = \binom{n}{r} p^r q^{n-r}$$

Diagram illustrating the components of the binomial probability formula:

- $\binom{n}{r}$ : # of trials (indicated by an arrow from "# of trials" to the  $n$ )
- $r$ : # of successes (indicated by an arrow from "# of successes" to the  $r$ )
- $p^r$ : prob. of success (indicated by an arrow from "prob. of success" to the  $p$ )
- $q^{n-r}$ : prob. of failure (indicated by an arrow from "prob. of failure" to the  $q$ )

**Ex 1.** If the probability that Paly will beat Gunn in a baseball game is 0.54, what is the probability that Paly will win exactly 2 out of 3 games in a series vs Gunn? Round to the nearest thousandth.

Does this have a fixed number of trials?  $\checkmark$  3

Is each trial independent?  $\checkmark$

Does each trial have only 2 possible outcomes?  $\checkmark$

Is the probability of success the same for each trial?  $\checkmark$  .55

What's my:  $n = 3$     $r = 2$     $p = .55$     $q = 1 - .55 = .45$

$$\begin{aligned} P(\text{win } 2) &= {}_3C_2 (.55^2) (.45^1) = \\ &= .408375 \\ &\approx .408 \end{aligned}$$

event flip coin

Ex 2. If a fair coin is tossed 3 times, find the probability of getting exactly one "heads". success = heads

Does this have a fixed number of trials? ✓

Is each trial independent? ✓

Does each trial have only 2 possible outcomes? -

Heads (success) Tails (fail)

Is the probability of success the same for each trial? → need to find p!

What's my:  $n = 3$   $r = 1$   $p = \frac{1}{2}$   $q = \frac{1}{2}$

$$P(\text{one head}) = {}_3C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2$$
$$= .1875$$
$$= .187 \text{ or } 18.7\%$$

\*ways to get what I want  
= total things that can happen  
← head  
 $= \frac{1}{2}$   
↑  
heads or tails

**You!** A quiz has 6 multiple-choice questions, each with 4 choices. If you guess at every question, what is the probability of answering 4 out of 6 questions correctly? Round to the nearest thousandth.

Does this have a fixed number of trials?

Is each trial independent?

Does each trial have only 2 possible outcomes?

Is the probability of success the same for each trial?

what's p? =  $\frac{1}{4}$

What's my:  $n = 6$   $r = 4$   $p = \frac{1}{4}$   $q = \frac{3}{4}$

$${}_6C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^2 = .03295$$
$$= .033$$
$$\text{or } 3.3\%$$

### What about questions with words like "at least" or "at most"?

**Ex. 3** If a fair coin is tossed 9 times what is the probability, rounded to the nearest thousandths, of getting at most 2 heads

$$p = \frac{1}{2} \quad q = \frac{1}{2}$$

Does this have a fixed number of trials?

Is each trial independent?

Does each trial have only 2 possible outcomes?

Is the probability of success the same for each trial?

$$n = 9$$

At most 2 heads out of 9 includes

$$\frac{0}{9}, \frac{1}{9} \text{ or } \frac{2}{9}$$

What are the ways I can get a success for at most two heads?

$$\begin{aligned} & \text{Zero out of 9} \quad r=0 \quad n=9 \quad + \quad \text{1 out of 9} \quad r=1 \quad n=9 \quad + \quad \text{2 out of 9} \quad r=2 \quad n=9 \\ & {}_9C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^9 \quad + \quad {}_9C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^8 \quad + \quad {}_9C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^7 \\ & .001953 \quad + \quad .017578 \quad + \quad .0703125 \\ & = .08984 = .090 \end{aligned}$$

**Ex 4.** Mr. and Mrs. Doran have a genetic history such that the probability that a child being born to them with a certain trait is 35%. If they have five children, what is the probability that at least two of their 5 children will have that trait? Round your answer to the nearest thousandths.

Does this have a fixed number of trials?

Is each trial independent?

Does each trial have only 2 possible outcomes?

Is the probability of success the same for each trial?

$$p = .35 \quad q = 1 - .35 = .65$$

$$n = 5$$

$$r = 2 \text{ or } 3 \text{ or } 4 \text{ or } 5$$

$$\begin{aligned} & r=2 \quad + \quad r=3 \quad + \quad r=4 \quad + \quad r=5 \\ & {}_5C_2 (.35)^2 (.65)^3 \quad + \quad {}_5C_3 (.35)^3 (.65)^2 \quad + \quad {}_5C_4 (.35)^4 (.65)^1 \quad + \quad {}_5C_5 (.35)^5 (.65)^0 \\ & (10) \quad (10) \quad (5) \quad 1 \\ & .33642 \quad .18115 \quad .04877 \quad .00525 \\ & = .5715 \\ & = .572 \end{aligned}$$

$$\begin{aligned} & {}_5C_1 (.35)^1 (.65)^4 \\ & 5 \cdot .312385 \\ & .68715 \end{aligned}$$

**You!** The Hiking Club plans to go camping in a State park where the probability of rain on any given day is  $\frac{3}{8}$ . What is the probability that it will rain at most one day out of the 7 days they are there? Round to the nearest thousandth.

Does this have a fixed number of trials?

Is each trial independent?

Does each trial have only 2 possible outcomes?

Is the probability of success the same for each trial?

Success = it rains

$$p = \frac{3}{8} = .375 \quad q = \frac{5}{8} = .625$$

$$n = 7 \quad r = 0 \text{ or } 1$$

$$r = 0 \quad + \quad r = 1$$

$${}_7C_0 (.375)^0 (.625)^7 + {}_7C_1 (.375)^1 (.625)^6$$

$$.037252 + .097789$$